



The extent to which the recommendations issued after the periodic evaluation of university programs are targeted, specific, measurable and unequivocal: An exploratory analysis

Chantal Bouchard^{a,*}, Marie-Claude Hamel^b

^a Department of Agri-Food Economics and Consumer Sciences, Université Laval, Québec, QC G1V 0A6, Canada

^b Analyste et conseillère en amélioration continue, Directeur des poursuites criminelles et pénales, Complexe Jules-Dallaire, tour 1, bureau 500, 2828 boulevard Laurier, Québec, QC G1V 0B9, Canada

ARTICLE INFO

Keywords:

Recommendation
Program evaluation
Quality
University-level teaching

ABSTRACT

Background and study objectives: Each year, Quebec universities conduct a periodic evaluation of their study programs and issue recommendations to improve the quality and relevance of these latter. Although these evaluations are commonplace in higher education institutions, there is little information about them and the resulting recommendations. To compensate this lack of convincing data, our exploratory analysis will look at the quality of expression of these recommendations.

Methodology: Experts were solicited to evaluate whether the recommendations were targeted, specific, measurable and unequivocal, four strategic characteristics of good formulations.

Results: Our results reveal major issues. Almost all of the recommendations do not indicate who is to carry them out. Also, the desired improvement is not clearly identified, thus leaving unanswered major questions about how they should be implemented. Another weak point was the absence of indicators for measuring the accomplishment of the recommendations. The last part of this study presents the best practices for issuing better recommendations, since these latter are key in the improvement process and the development of action plans. This study opens the way for research that will examine the root causes of the problem and make it possible to establish best practices for writing recommendations.

1. Introduction

Quality assurance in higher education institutions and of their study programs first appeared in the early 1990s (Kanji et al., 1999). This permanent and ongoing process intended to evaluate and improve the quality of establishments and/or their programs is based on highly defined criteria (Martin & Ouellet, 2012, p. 5).

While the literature dedicated to quality assurance (QA) in the field of education is lacking compared to other educational topics (Khuram et al., 2023), it would be a misjudgment to underestimate the inherent value of the constructive role played by higher education institutions (HEIs) in shaping society (Păunescu et al., 2022). A growing interest in quality assurance is becoming evident highlighted by the research conducted by (Sezgin et al., 2022).

It is important to emphasize that there remain unexplored research avenues aimed at thoroughly investigating the societal impact of higher

education programs. Among these avenues of investigation, the analysis of how quality assurance measures distinctly influence this impact carries significant importance.

2. Quality assurance systems

There are many quality assurance systems, and they vary from one country to the next. France, the United States, and Great Britain each have independent national evaluating agencies. These agencies operate independently from the government and universities. Their role is to monitor and attest to the quality of training programs. (Conférence des recteurs et des principaux des universités du Québec, 2004). In Canada where education is the responsibility of the provinces and territories, various quality assessment practices have been instituted and nine provinces are equipped with such organizations (Corbo, 2013).

In Quebec, each university establishes its own system for evaluating

* Corresponding author.

E-mail address: chantal.bouchard@fsaa.ulaval.ca (C. Bouchard).

<https://doi.org/10.1016/j.evalprogplan.2023.102364>

Received 28 November 2022; Received in revised form 16 August 2023; Accepted 20 August 2023

Available online 21 August 2023

0149-7189/© 2023 Elsevier Ltd. All rights reserved.

academic programs, which is overseen by the *Policy of Québec Universities on the Periodic Evaluation of Current Academic Programs*. This policy sets standards and minimum conditions that universities must meet according to their own institutional policies. (*Conférence des recteurs et des principaux des universités du Québec, 2004*).

2.1. Description of the periodic evaluation process

Every ten years, each degree program undergoes a thorough examination process that determines its quality and relevance. This process defines the context in which the program exists and evolves, and highlights its characteristics, strengths and weaknesses. In the goal of objectivity, this process is comprised of three perspectives: internal, external and institutional (*Conférence des recteurs et des principaux des universités du Québec, 2004*).

The first perspective uses self-evaluation by an external committee of professors, lecturers and students that writes a report describing the program, its strengths and weaknesses, and issues recommendations for its improvement. When needed, a development project proposal is provided, following a rigorous process (*Conférence des recteurs et des principaux des universités du Québec, 2004*).

From the process' external perspective, experts recognized in the field under evaluation take stock of the program. A group of evaluators, including at least one university professor, is responsible for assessing the program. They conduct a self-evaluation and visit the establishment, and then issue a final judgement on the program's quality, relevance, and the measures that can be taken to improve it. (*Conférence des recteurs et des principaux des universités du Québec, 2004*).

The institutional perspective is implemented by an institutional committee of academics and professors who are not involved in the program under evaluation. The committee critically analyzes the self-evaluation report and the experts' advice, summarizing for university authorities the program's strengths and weaknesses and any needed improvements. Once the committee's role is fulfilled, its report provides elements needed for decision making and for writing a summary of the evaluation (*Conférence des recteurs et des principaux des universités du Québec, 2004*).

Finally, a decision-making body takes position on the recommendations laid out in the report and the action plan (*Conférence des recteurs et des principaux des universités du Québec, 2004*). The process ends with the internal and external dissemination of an evaluation summary to meet accountability criteria. This informs the university community and society of the state of the program and the measures selected to improve its quality (*Conférence des recteurs et des principaux des universités du Québec, 2004*). Fig. 1 illustrates the process of the periodic evaluation of Quebec university programs.

2.1.1. Observations of para-university players

Although important, the periodic evaluation is cause for concern since, at times, it is performed simply to meet administrative requirements. Some problems arise: the unwieldiness of voluminous data collection, the lack of resources for analysis, the number of people required, the necessary qualifications, and the costs and length of the evaluation (*Bernatchez, 2009*). Little information is provided about these periodic evaluations and the implemented improvements (*Fédération du personnel professionnel des universités et de la recherche, 2011*).

The Program Evaluation Review Commission (the Commission de vérification de l'évaluation des programmes, or the CVEP) is designated to verify institutional practices. In keeping with its mandate, the CVEP confirmed the conformity of universities' institutional procedures with the CREPUQ's policy and did so by questioning its pertinence (2008, pp. 13–16). It was unable to establish if the evaluation had real effect on the improvement of the program. There was no information on the impact of the evaluations, on the application of the recommendations that had been formulated or on the implementation of action plans (2008;

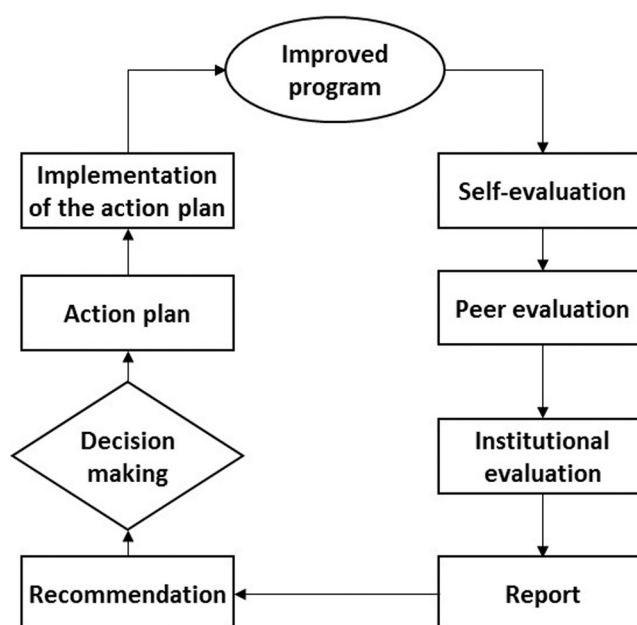


Fig. 1. Process of the periodic evaluation of Quebec university programs.

Conférence des recteurs et des principaux des universités du Québec, 2004). Within its broad mandate, the CVEP sought to know if all programs had been evaluated and if the establishments followed up as planned on the periodic evaluations. The CVEP established a first overview on the recommendations issued in the evaluations and then assessed their level of application, highlighting the variations from one university to the next. It showed that some recommendations were yet to be implemented five years later, while others had been abandoned (2013, p. 36). Several justifications for this tardiness were given: lack of resources, hesitation, the variable length of the implementation and the difficulty of assigning the responsibility of the implementation. The recommendations' lack of precision, their unrealistic nature and non-feasibility add to the list of reasons the recommendations were abandoned (2013, p. 32), leading the CVEP (2013, p. 34) to emit two recommendations:

1. It is required that establishments formulate recommendations stemming from periodic evaluations in a specific and operational way, ensuring their feasibility to facilitate follow-up.
2. That, in so doing, however, the establishments do not unduly disregard innovative recommendations just because they are complex to implement.

2.2. Objectives of the study

The observations of university stakeholders raise questions about universities' internal quality assurance mechanisms. Periodic program evaluation requires substantial human and financial resources, but little information is provided about these evaluations and their real impact on the improvement of programs. The CVEP provided us with information on the follow-up of evaluations, which revealed troubling observations. This information revealed inadequate improvement processes and varying implementation rates across universities. The information also showed that some recommendations remained unimplemented after five years, while others were abandoned because they were not clear, feasible, or realistic.

The only available information on these recommendations comes from the CVEP. The strategic nature of this information will allow this study to fill in informational gaps in this part of the evaluation process. Few empirical studies have been conducted on these recommendations,

thus justifying the pertinence of this analysis. Scholars interested in this issue believe that it is time to increase the body of knowledge on the subject (Patton, 1997). This study will take stock of current recommendations in the periodic program evaluation reports. It will seek to answer the following question: How well are the issued recommendations expressed in the periodic program evaluations? Since there is little information currently available on the periodic evaluation process and on its real impact on improving programs, this study will certainly contribute to the literature.

3. Evaluation framework

The concept of recommendations has no universal definition. There is, however, a set of terms: hypothesis on the future (Patton, 1997, p. 328), suggestions and optional status (Iriti et al., 2005; Patton, 1982), and decision-making help (Fitzpatrick et al., 2011).

3.1. Characteristics of the recommendations

To be useful to decision makers, recommendations must have two categories of qualities: extrinsic and intrinsic.

3.1.1. Extrinsic qualities

Extrinsic qualities ensure the validity of the recommendations. A good recommendation must be *relevant*. Logically stemming from evaluation results (Fitzpatrick et al., 2011), it cannot go beyond the data and remains rooted in the evaluation results (Torres et al., 1997). It offers answers to initially asked questions (Fitzpatrick et al., 2011, p. 474) that are in keeping with the organization's interests (Diallo, 2011, p. 42).

Another key quality is the *justification* of the recommendation (Grosshans, 1994, p. 11; Patton, 1997, p. 325). This quality is based on a set of facts and proof gathered during the evaluation process. The evaluator will use this information as the basis for their recommendations. (Gibbs & Maguire, 2012, pp. 472–474). *Feasibility* is another of the main qualities of a good recommendation (Joras, 2000) and reflects the facility of implementing the recommendation (Handley & Green, 2004). A meta-quality according to Mitchell (2009) and Joras (2000), *acceptation* depends on the other qualities to be effective. Acceptation of the recommendations by decision makers does not guarantee results (Grosshans, 1994, p. 1); in fact, it is only the starting point for tangible actions that will lead to improving the program.

3.1.2. Intrinsic qualities

Intrinsic qualities, which are more suited to addressing formulation, are essential for an effective communication of the improvements and the acceptance of the recommendations (Gibbs & Maguire, 2012). Many authors stress the importance of correctly formulating recommendations (Centre Européen d'Expertise et d'Évaluation, 2014; Grosshans, 1991, p. 10; Handley & Green, 2004). When well-targeted, recommendations solicit the stakeholders in charge of and holding the power to implement them (Centre Européen d'Expertise et d'Évaluation, 2014; Grosshans, 1991, p. 10; Handley & Green, 2004, p. 10). This process favors player's contribution to the improvement process (Centre Européen d'Expertise et d'Évaluation, 2014; Handley & Green, 2004) and follow-up (2014).

Several authors have paid particular attention to a recommendation's ideal degree of *specificity*. Various positions have been taken, advocating for a range from the most general to the most specific of formulations. Some prefer recommendations that very specifically (Brown et al., 2006; Handley & Green, 2004; Patton, 1982; Sharen, 2014; The Hong Kong Polytechnic University, 2013) indicate the desired improvement (Patton, 1982, pp. 109, 287). They must express just one idea for improvement (Handley & Green, 2004; Sharen, 2014; The Hong Kong Polytechnic University, 2013) and set out the actions that must be taken to bring it about (Brandon, 1999, p. 42; Fitzpatrick et al., 2011, p. 475; The Hong Kong Polytechnic University, 2013, p. 3). Other authors stress the need for the recommendation to be *measurable*. Written in

quantifiable and tangible terms (De Waart, 2006; Handley & Green, 2004; Sharen, 2014), a measurable recommendation favours stakeholders' follow-through (Gibbs & Maguire, 2012), its evaluation and the extent of its implementation (McLoughney, 2007; The Hong Kong Polytechnic University, 2013).

Authors offering advice on formulating recommendations agree on the importance of being clear and consistent (Bouin & Simon, 2000, p. 230; Fitzpatrick et al., 2011, p. 464; Patton, 1997; Sattler, 2001, p. 689; The Hong Kong Polytechnic University, 2013, p. 6). *Unequivocal* writing leads to just one interpretation and therefore effectively communicates content (Fitzpatrick et al., 2011). Clarity makes an author credible (Bouin & Simon, 2000), allows for a better understanding of the recommendations and increases the use of the evaluation report (Bouin & Simon, 2000).

4. Qualities retained for the study

While it would have been interesting to evaluate all of the listed qualities, choices had to be made. Since several intrinsic qualities (relevance, feasibility, etc.) can only be understood with full analysis of the evaluation reports or an internal analysis of the program, we only retained extrinsic qualities. In keeping with our study's objective, Fig. 2 shows a schematic of the definition of a recommendation and the qualities assessed.

4.1. Methodology

4.1.1. Creation of a bank of recommendations

The bank of recommendations was constituted in a non-probabilistic manner, from a sampling of volunteers (Dépelteau, 2000, p. 229). To obtain the latest periodic program evaluation reports, we approach all Quebec universities. The rules for the inclusion of these reports were:

1. Last periodic evaluation completed
2. Undergraduate program of study offered

Two-hundred and twenty-two recommendations were extracted from 24 reports from the 10 universities that accepted to participate in the project. To protect participants' identity, minimize bias errors and facilitate reading, the names of the universities, departments and study programs were removed. The course numbers and certain specific terms were replaced by "xxx." Six recommendations were eliminated, due to a loss of their original meaning.

4.1.2. Recruitment of experts

Two distinct groups of experts were solicited. The first (Group 1) was made up of external evaluation experts from the university, public and private sectors. Internal experts, faculty deans and program directors made up the other group (Group 2). The sampling procedure for volunteers was retained and combined with "snowball sampling" for Group 1 (Dépelteau, 2000, p. 227). In the end, 58 experts accepted to participate in the study. Thirty-six were evaluation experts, 12 were program directors and 10 were deans.

5. Evaluation tool

An analytical and four-scale description evaluation grid was developed, making it possible to collect feedback on the observed strengths and weaknesses of each evaluated recommendation (Scallion, 2004, p. 182). The use of the descriptive scales had the advantage of standardizing evaluators' feedback, thus improving adherence and precision (Côté & Tardif, 2011, p. 25). To maintain the measurement instrument's user-friendliness, each descriptive scale had three levels and space for comments. Two recognized evaluation experts validated the semantics of the grid, leading to minor modifications.

To considerably reduce the number of recommendations each expert

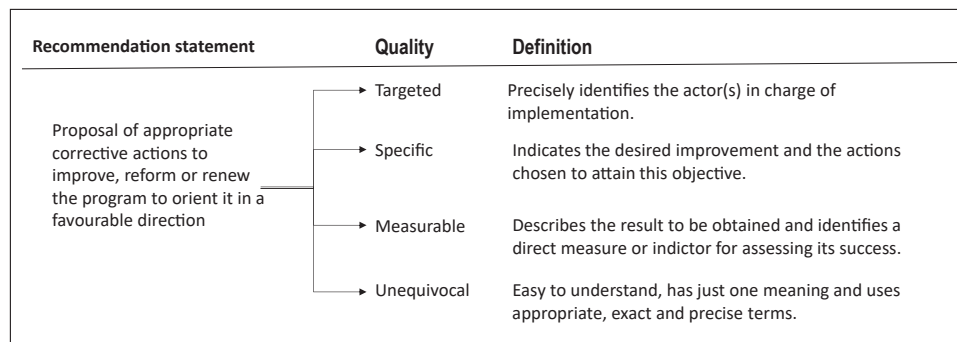


Fig. 2. Framework for evaluating the qualities of the recommendations' formulation.

evaluated, a random drawing was held from the bank of 222 recommendations to distribute 22 or 23 statements into ten distinct versions of the tool. A lesser number of recommendations lightened the participants' workload and increased both the response rate and the quality of the responses (Galesic, 2002; Herzog & Bachman, 1981).

6. Results

The following section presents a global analysis of the results, laid out by the recommendations' quality and qualitative nature. We used IBM-SPSS software to process our data.

7. Global analysis

This analysis of recommendations offers an overview of the results by presenting the agreement of the experts' evaluations, which we measured using the intraclass correlation coefficient (ICC). The nature of the measurement scale and the necessity of finding agreement between multiple experts led us to choose ICC over the weighted kappa. The weighted kappa is limited to the calculation of agreement between two experts. The ICC emphasizes the similarity of the evaluations and their severity. Varying between -1 and $+1$, a co-efficient close to 1 indicates a nearly perfect agreement between experts, while a co-efficient of 0 shows a random agreement. For negative values, the agreement between experts was poor (Hallgren, 2012). The Cicchetti (1994) were used to interpret the CCI. The values of 0.75–1.00 indicate an "excellent" agreement; from 0.60 to 0.74 a "good" agreement; from 0.40 to 0.59 a "fair" agreement; 0.39 and under a "poor" agreement.

Table 1 displays the distribution of respondents, the inter-expert agreement for each questionnaire, the qualities being evaluated and the interpretation of ICC according to Cicchetti (1994). Missing values made up about 5% of our data, and therefore did not alter the validity of the study's results (Giorgi, 2014). In this case, the ICC was calculated on

a lesser number of recommendations. For 6 questionnaires out of 10 (B-C-E-H-I-J), the experts' agreement was good for most of the qualities being evaluated. The four other questionnaires (A-D-F-G) showed fair agreement between experts, thus indicating their diverging opinions. The majority of questionnaires obtained excellent agreement. This is a solid basis for analyzing the quality of the recommendations.

The experts completed 67% of the questionnaires (39 out of 58). This proportion was higher among external (75%; 26/36) than internal experts (55%; 12/22).

Table 2
Distribution of expert answers, by quality.

Evaluated quality	Scale		
	The quality is absent/non-identifiable, n/N (%)	The quality is implicitly presented, unclear or lacking, n/N (%)	The quality is presented explicitly and easily recognizable, n/N (%)
Targeted (N = 178, 44 unclassifiable)	118/178 (66%)	55/178 (31%)	5/178 (3%)
Specific (N = 145, 77 unclassifiable)	10/145 (7%)	64/145 (44%)	71/145 (49%)
Measurable (N = 141, 81 unclassifiable)	60/141 (43%)	38/141 (27%)	43/141 (30%)
Unequivocal (N = 156, 66 unclassifiable)	8/156 (5%)	53/156 (34%)	95/156 (61%)

Table 1
Distribution of respondents and interexpert agreement.

Questionnaire	Proportion of internal experts*, n/N (%)	Targeted, CCI (n)	Specific CCI (n)	Measurable CCI (n)	Unequivocal, CCI (n)
A	2/4 (50%)	0.70 (23)	Good	0.34 (22)	Poor
B	1/3 (33%)	0.26 (22)	Poor	0.70 (22)	Good
C	1/4 (25%)	0.69 (22)	Good	0.55 (22)	Fair
D	1/3 (25%)	0.69 (19)	Good	0.18 (19)	Poor
E	1/3 (25%)	0.72 (22)	Good	0.39 (20)	Poor
F	0/4 (0%)	0.13 (20)	Poor	0.44 (20)	Fair
G	1/5 (20%)	0.48 (22)	Fair	0.35 (22)	Poor
H	2/4 (50%)	0.88 (21)	Excellent	0.49 (21)	Fair
I	2/4 (50%)	0.78 (22)	Excellent	0.54 (22)	Fair
J	1/5 (20%)	0.72 (20)	Good	0.71 (20)	Good
Total	12/39 (31%)	0.61 (213†)	Good	0.47 (210†)	Fair

* Internal experts are deans (n = 6) and program directors (n = 6). External experts come from university (n = 8), public (n = 10) and private (n = 9) sectors.

† The proportion of missing answers was 5% (4% for Targeted, 5% for Specific, 6% for Measurable and 6% for Unequivocal).

7.1. Analysis by quality

A detailed analysis for the four qualities was conducted. Table 2 shows the distribution of experts' responses.

The positioning of the recommendations on the three-level descriptive scale confirms poor results for the "targeted" and "measurable" qualities. The stakeholder in charge of implementing the recommendation can be identified only 33% of the time, but clearly only 3% of the time. In 43% of cases, the recommendation does not specify a measure/indicator for assessing its implementation. The "specific" and "unequivocal" qualities obtain better results. The recommendations are mostly unequivocal (61%) and specific (40%), while between 20% and 35% cannot be classified for evaluation.

7.2. Qualitative analysis

A qualitative analysis of the recommendations was carried out to account for the statements that were well or not well formulated and to show the general trends for each quality. This is displayed in Table 3.

The recommendations were deemed to be "targeted" when the type of stakeholder implementing them is clearly identified; if it is not clear, these recommendations are not considered to be targeted. As one expert said, the use of the terms like "school" or "the program" are "... so broad no one feels stirred to action."

"Specific" recommendations clearly present the desired improvement and set out a process to attain it, but they lack precision. Experts pointed out that "the expected action is described, but without any mention of the desired improvement," or that "the meaning of the improvements is understood, it is not specified enough to orient the action."

According to the experts' perceptions, "measurable actions" show a desire result and the "countable" means to measure it. When they are less measureable, it is difficult to find a precise indication of the changes to be made, the expected results are unclear and use terms like "reinforce," "more," "better" and "balance."

The experts felt that unequivocal recommendations favour simple terms over the technical and difficult-to-understand academic terms. The recommendations that caused confusion used words like "better," "more," "pay special attention," "certain gaps," "if required," "make an effort" and "appropriate authorities." Without any further precision, this vocabulary reduces the recommendations' specificity, measurability and clarity.

8. Discussion

This study draws a portrait of the recommendations issued in the reports following periodic evaluation of Quebec university programs. In general, these recommendations are perceived by experts to be specific and unequivocal, though not always explicit. The majority of recommendations are not well targeted and their degree of measurability fluctuates greatly.

In this discussion, we will interpret these quantitative and qualitative results and issue a series of suggestions to help the writers of these reports.

8.1. Targeted

Our results show that the infrequent identification of the stakeholders in charge of implementing the recommendations is the primary weakness of periodic evaluation reports. It is usually entities that are named for implementation, yet it is rather soliciting responsible stakeholders that favours their adherence to change (Centre Européen d'Expertise et d'Évaluation, 2014; Handley & Green, 2004).

An explanation of these results is the fact that stakeholders produce these evaluations in each program. It would seem that the

Table 3

Examples of well and less-well formulated recommendations, by quality.

	Well-formulated statements	Poorly formulated statements
Targeted	"That the <u>Program Committee</u> analyses the content of xxx courses to confirm students attain the desired objectives." "That the <u>department's professors, lecturers and students</u> undertake a strategic planning process for the whole department that takes into account undergraduate students' needs for training." "That the <u>School</u> produces a milestone report before June 2016 on the increase in internship hours to meet the 700-hour standard." "Call on the xxx Service to review the educational approach used in distance learning courses." "That the <u>School</u> adopts a policy on the professional aptitudes students require."	"Establish and promote a targeted strategy for marketing the program." "Further the reflection on requiring three intervention methods in internships and make the necessary adjustments." "Differentiate and define expectations for internships in the <u>Internship Guide</u>." "Find ways to improve the graduation rate." "Standardize the quality of different cohorts' training."
Specific	"Create a <u>committee to develop new learning methods</u> in business workplaces." "Using contests held by the appropriate authorities, set up a program for recruiting very good students outside of Quebec." "That the reasons student drop out of the undergraduate program be identified and means be found to <u>improve graduation rates</u>." "Based on TECFEE's example for the quality of French language skills, <u>evaluate acquired math skills</u> using a <u>diagnostic test</u>. Develop this test in collaboration with the xxx department. According to test results, <u>orient students toward refresher courses</u> (without additional credits)." "That the <u>School</u> produces milestone report before June 2016 on the increase in internship hours to meet the 700-hour standard."	"Reflect on the role of the two <u>research classes, given future certification requirements</u>." "That appropriate authorities <u>reinforce</u> the coherency and balance of training for various paths." "Clarify and maintain the role of theology and religion in the program and ensure their <u>usefulness and reciprocity</u>." "Reorganize and trim down the <u>current program</u> to set up a <u>new structure</u>." "Be diligent about BSW by stating what will become of this <u>program in the report to be submitted in April 2012</u>."
Measurable	"Ensure a <u>parity between teachers and students</u> on the program committee." "<u>Modify the content in the course outline for the xxx class</u> by adding a mention that this course has aspects that may have been taught in the required courses for students in the xxx undergraduate program." "Offer an <u>internship possibility in each concentration</u>." "To harmonize the course offering shared with the certificate program following its recent modifications, <u>replace the xxx course by the xxx course in the pool of courses</u>." "Increase the presence of <u>supervisors</u> in internship workplaces."	"Strike a balance between the discipline's theoretical aspects and professional practice." "Collect more data showing the development of the 12 skills in class, the activities and internships to ensure the programs' ongoing improvement. If necessary, improve course outlines in consequence." "Reinforce sports education." "That undergraduates in work-study programs be better prepared." "That possible solutions to plagiarism and to administration of tests be seriously examined."

(continued on next page)

Table 3 (continued)

	Well-formulated statements	Poorly formulated statements
Unequivocal	“Make printed (books and specialized magazines) and online resources available.” “Outline the skills acquired in the program and make them known to students and potential employers.” “Revisit internship structure and modalities to improve both progression through the activities and concertation between the different stakeholders, in such a way that: - more time, and therefore importance, is given to Stage I (at the moment, it is only a one-week observation internship); - the number of supervisor visits are increased in Stage IV; - the supervision and evaluation requirements are standardized between the various supervisors for the same internship.” “Guarantee a minimum of two professors per concentration and per campus for undergraduates and three per concentration if a Master’s program is also offered.” “Expand the selection of host establishments for students who wish to study abroad.” “Make printed (books and specialized magazines) and online resources available.”	“Reinforce sports education.” “Work to make connections and build bridges between teaching elements.” “Pursue, if so required, the revision of the outlines for required courses with an eye to the professional skills presented in each activity and the evaluation carried out to ensure their mastery.” “Set up internal program review mechanisms that allow for a periodic or even regular re-evaluation of relevance and usefulness.” “That partnerships judged to be useful with the best CEGEPs be set up or strengthened.” “Reinforce sports education.”

recommendations are intended for the program’s highest manager. This method was favoured by one of the study’s experts, who implicitly addressed the recommendations to the organization leader. This is a worthwhile method if the evaluator has limited knowledge of the organization being assessed. However, it is not useful in the periodic evaluation of university programs when the evaluator participates in the field. For clarity’s sake, it is important to precisely designate an individual with the authority and power to implement the recommendations and to make this stakeholder accountable for doing so.

8.2. Specific

The “specific” quality did not produce very good inter-expert agreement, perhaps because of the complexity of evaluating this quality. Positioning the recommendations on the descriptive scale showed that a little over a third of the recommendations did not reach agreement with a majority of experts, another third of the recommendations were expressed imprecisely and the last third clearly indicated the desired improvement.

When examining the current formulations, experts felt the recommendations were very specialized. According to one participant, some recommendations were “more of a solution to an unnamed problem than a recommendation [...] Here, it would seem that those issuing the recommendations are soliciting decision-makers’ belief rather than their judgment.”

It would be useful to examine exactly what level of specificity is needed for the recommendations. According to Iriti et al. (2005), this quality varies according to the evaluator’s level of organizational knowledge. When evaluators have extensive organizational knowledge, as is the case in the periodic evaluation, they are able to suggest specific options and ways of implementing these options.

This does not appear to be systematic since the desired improvement was not always clearly articulated. To improve the way

recommendations are formulated, three key steps are necessary: a clear answer is needed, the effect that is sought from the change must be made known and the specific actions have to be suggested to improve the program. By integrating these three elements, a full improvement strategy is expressed and the suggestion of specific actions shows that the recommendation can indeed be implemented (Gibbs & Maguire, 2012).

8.3. Measureable

Nearly half of the recommendations were formulated in such a way that makes the objective measurement and evaluation of their implementation difficult or even impossible. The recommendations must use quantifiable elements to clearly define the objectives, limit the potential for interpretation and obtain desired results. A precisely expressed result allows those targeted by the recommendation to grasp the extent of the change and assess its implementation. According to one expert, the question should be, “What [is the] threshold that must be reached to call the implementation of the recommendation a success?” When a recommendation includes a target that can be used to evaluate whether the desired outcome has been achieved, it becomes possible to establish a direct measure. This can improve both the recommendation’s measurability and specificity.

8.4. Unequivocal

The heterogeneous nature of evaluations can be explained by the diversity of the experts who took part in this study, many of whom were from beyond the university setting. As one expert said, what is unequivocal in one field is not necessarily so in another.

The results show that many of the evaluated recommendations were formulated adequately and in such a way as to articulate a clear and easy-to-understand message. When university-specific terms were used, however, understanding becomes more difficult. Some words were seen as causing confusion and reducing the specificity, measurability and clarity of the recommendations. As Fitzpatrick et al. (Fitzpatrick et al., 2011) pointed out, technical terms and jargon should be avoided, except when their use ensures the report’s clarity and credibility. Jargon is justified when recommendations are expressed and used among university staff (since this language contributes to the report’s clarity), but its use must remain limited.

Recommendations should present clear ideas that express expectations precisely and leave no room for interpretation. This would ensure that the results obtained meet original expectations. Ambiguous formulations can affect the implementation of recommendations and, ultimately, the improvement brought to the program.

9. Conclusion

This study sought to draw a portrait of the recommendations stemming from the periodic evaluations of Quebec university programs. The little information available and the strategic nature of the recommendations, which are the basis of improvements to the higher-learning programs, motivated this study. Positive response from many evaluation experts confirmed the interest in continuing to develop this knowledge.

The participation of experts from various settings allowed us to evaluate how well recommendations were formulated and whether they were targeted, specific, measurable and unequivocal. Among the issues observed, we saw that the expression does not formally identify the stakeholders responsible for implementing the recommendation. The improvement being targeted by the recommendations is not always mentioned clearly and, in fact, often only the specific actions are suggested. The expression’s qualitative nature makes it difficult to objectively measure its implementation. Many recommendations are adequately unequivocal. These observations made is possible to propose

best practices for formulating recommendations.

Focusing on the intrinsic qualities of how the recommendations were formulated, this study looked at a fundamental aspect that neglects extrinsic qualities, which are important in the recommendations' quality assurance and improvement process. Further research on the evaluation process will be necessary to examine the recommendations' extrinsic qualities. Experts were consulted only once for this study. There was no consensus and several recommendations remain without a definitive answer as to their quality. A second round of consultations would have drawn a more precise portrait and reduced the number of unclassified recommendations. The DELPHI technique could have been used to obtain expert agreement on all recommendations.

To date, few studies have been conducted on this subject and, therefore, there are many possibilities yet to explore. The results obtained and the suggestion of best practices could serve as the basis for developing a diagnostic tool for improving the quality of the formulation of recommendations. This tool would be applicable in university settings to improve the evaluation process there and therefore make sure evaluators formulate higher quality recommendations. The logical next step to this study would be to establish links between the quality of the formulation of the recommendations and their level of implementation.

Author statement

We, the authors of this manuscript, have reviewed and approved the final version of the article being submitted. We confirm that this article is our original work, and it has not been previously published nor is it under consideration for publication elsewhere. By signing this author agreement, we warrant that the article meets the ethical standards and guidelines for publication in the respective journal."

References

- Bernatchez, J. (2009). Principes, modalités et enjeux de l'évaluation des activités des professeurs d'université au Québec. *Questions Vives Recherches Éno'sis éducation*, 6 (12), 13–27.
- Bouin, X., & Simon, F.-X. (2000). *Les nouveaux visages du contrôle de gestion*. Dunod. (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=i2100048872>).
- Brandon, P. R. (1999). Involving program stakeholders in reviews of evaluators' recommendations for program revisions. *Evaluation and Program Planning*, 22(3), 363–372. [https://doi.org/10.1016/S0149-7189\(99\)00030-0](https://doi.org/10.1016/S0149-7189(99)00030-0)
- Brown, P., Brunnhuber, K., Chalkidou, K., Chalmers, I., Clarke, M., Fenton, M., Forbes, C., Glanville, J., Hicks, N. J., Moody, J., Twaddle, S., Timimi, H., & Young, P. (2006). How to formulate research recommendations. *BMJ*, 333(7572), 804–806. <https://doi.org/10.1136/bmj.38987.492014.94>
- Centre Européen d'Expertise et d'Évaluation. (2014). *Boîte à outils*. Société Eureval. Retrieved 10–04 from (http://www.eureval.fr/-Les-recommandations-d-une.html?var_recherche=recommandation).
- Cicchetti, D. V. (1994). Guidelines, criteria, and rules of thumb for evaluating normed and standardized assessment instruments in psychology. *Psychological Assessment*, 6 (4), 284.
- Conférence des recteurs et des principaux des universités du Québec. (2004). Guide d'application de la politique de la CREPUQ relative à l'évaluation périodique des programmes existants.
- Corbo, C. (2013). *Rapport du Chantier sur un Conseil national des universités*. (http://www.mesrst.gouv.qc.ca/fileadmin/administration/librairies/documents/chantiers/RapportChantier_CNU_RapportCorbo.pdf).
- Côté, R., & Tardif, J. (2011). Élaboration d'une grille d'évaluation. *Québec: Document académique*.
- De Waart, D. (2006). GETTING SmART. *Supply chain Management Review*, 27.
- Dépelleau, F. (2000). *La démarche d'une recherche en sciences humaines*. De Boeck Supérieur.
- Diallo, N. (2011). *La valorisation de l'utilisation des évaluations de programme: cas de la Banque mondiale au Burkina Faso* Université Laval.
- Fédération du personnel professionnel des universités et de la recherche. (2011). Assurance qualité au Québec dans la formation universitaire., In (pp. 12). Trois-Rivières (Québec): La Fédération.,
- Fitzpatrick, J. L., Sanders, J. R., & Worthen, B. R. (2011). *Program evaluation: alternative approaches and practical guidelines* (fourth ed.). Pearson Education (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=a2004157>).
- Galesic, M. (2002). *Effects of questionnaire Length on Response rates*.
- Gibbs, P., & Maguire, K. (2012). What is in a recommendation? A perspective from work-based doctorates. *Research in Post-Compulsory Education*, 17(4), 471–481. <https://doi.org/10.1080/13596748.2012.738994>
- Giorgi, R. (2014). *Traitement des données manquantes*. Retrieved 29–07 from (http://cyberim.timone.univ-mrs.fr/Members/rgiorgi/DossierPublic/Enseignement/TraitementNA-PDF-RG/docpeda_fichier).
- Grosshans, W. (1991). How to Get Action on Audit Recommendations. *The Government Accountants Journal*, 43(1), 17.
- Grosshans, W. (1994). How to Get Action on Audit Recommendations. *The Government Accountants Journal*, 43(1), 17.
- Hallgren, K. A. (2012). Computing inter-rater reliability for observational data: An overview and tutorial. *Tutorials in Quantitative Methods for Psychology*, 8(1), 23.
- Handley, M., & Green, R. (2004). *Safeguarding through audit A guide to auditing case review recommendations*. National Society for the Prevention of Cruelty to Children. (<http://www.nspcc.org.uk/globalassets/documents/advice-and-info/safeguarding-through-audit.pdf>).
- Herzog, A. R., & Bachman, J. G. (1981). Effects of questionnaire length on response quality. *Public Opinion Quarterly*, 45(4), 549–559.
- Iriti, J. E., Bickel, W. E., & Nelson, C. A. (2005). Using recommendations in evaluation: A decision-making framework for evaluators. *American Journal of Evaluation*, 26(4), 464–479. <https://doi.org/10.1177/1098214005281444>
- Joras, M. (2000). *Les fondamentaux de l'audit* (econd ed.). Ed. Préventique (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=01-0595701>).
- Kanji, G. K., Malek, A., & Tambi, B. A. (1999). Total quality management in UK higher education institutions [Article]. *Total Quality Management*, 10(1), 129–153. <https://doi.org/10.1080/0954412998126>
- Khuram, S., Rehman, C. A., Nasir, N., & Elahi, N. S. (2023). A bibliometric analysis of quality assurance in higher education institutions: Implications for assessing university's societal impact. *Evaluation and Program Planning*, 99. <https://doi.org/10.1016/j.evalprogplan.2023.102319>
- Martin, E., & Ouellet, M. (2012). *Les mécanismes d'assurance qualité dans l'enseignement supérieur*. I. d. l. r. e. d. i. socio-économiques. (<http://www.iris-recherche.qc.ca/wp-content/uploads/2012/11/Assurance-qualite-web.pdf>).
- McLoughney, S. (2007). Making the most of performance review: With SMARTER goals and best feedback. *Accountancy Ireland*, 39(6), 56–58. (<https://ezproxy.bibl.ulaval.ca/login?url=https://search.proquest.com/docview/223184520?accountid=12008>).
- Mitchell, R.-C. (2009). *Strategy Formulation*. Retrieved 10–04 from (<http://www.csun.edu/~hfmgt001/writing.htm>).
- Patton, M. Q. (1982). *Practical evaluation*. Sage Publications., (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=01-1434993>).
- Patton, M. Q. (1997). *Utilization-focused evaluation: the new century text* (third ed.). Sage Publications (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=i9780803952652>).
- Păunescu, C., Nikina-Ruohonen, A., & Stukalina, Y. (2022). Fostering research with societal impact in higher education institutions: A review and conceptualization. In C. Păunescu, K.-L. Lepik, & N. Spencer (Eds.), *Social Innovation in Higher Education: Landscape, Practices, and Opportunities* (pp. 153–178). Springer International Publishing. https://doi.org/10.1007/978-3-030-84044-0_8.
- Sattler, J. M. (2001). *Assessment of children: cognitive applications* (fourth ed.). Jerome M. Sattler, Publisher, Inc.. <http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=01-0601560>.
- Scallon, G. (2004). *L'évaluation des apprentissages dans une approche par compétences*. Éditions du Renouveau pédagogique Inc. (<http://ariane.ulaval.ca/cgi-bin/recherche.cgi?qu=a1408939>).
- Sezgin, A., Orbay, K., & Orbay, M. (2022). *Educational Research Review From Diverse Perspectives: A Bibliometric Analysis of Web of Science (2011–2020)*, 12. SAGE Open. <https://doi.org/10.1177/21582440221141628>
- Sharen, C. (2014). Is-SMART Recommendations. (<http://colleensharen.wordpress.com/2014/04/15/is-smart-recommendations/>).
- The Hong Kong Polytechnic University. (2013). *Report Writing: Recommendations*. Retrieved 12–04 from http://www2.elc.polyu.edu.hk/cill/eiw/report_recommendations.htm.
- Torres, R. T., Preskill, H. S., & Piontek, M. E. (1997). Communicating and reporting: Practices and concerns of internal and external evaluators. *American Journal of Evaluation*, 18(2), 105–125. <https://doi.org/10.1177/109821409701800203>

Chantal Bouchard, Associate Professor at Laval University, specializes in the field of measurement, evaluation and quality management as well as designing executive improvement programs and manager training. Professor Bouchard holds a PhD in Measurement and Evaluation, and her research focuses on developing best practices for quality management in various industries and has been consultant in Quality Management for nearly two decades.

Marie-Claude Hamel, analyst and advisor in continuous improvement to the Directeur des poursuites criminelles et pénales, specializes in the development of management information and the improvement of business processes. The analyst holds a master's degree in consumer science and a Lean Six sigma Green Belt certification.